INTERIM REPORT

A Statistical Approach to Monitor Quantisation in Neural Network Training

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PLÁN OBNOVY
ÚRAD PODPREDSEDU VLÁDY SLOVENSKEJ REPUBLIKY PRE PLÁN OBNOVY A ZNALOSTNÚ EKONOMIKU
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„A Statistical Approach to Monitor Quantisation in Neural Network Training"

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Project name: A Statistical Approach to Monitor Quantisation in Neural Network Training

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PROJECT METHODOLOGY

The methodology of the project is structured into four follow-up phases (Milestones), where individual tasks are incorporated in the form of Work Packages (WP) that comprehensively address the challenges of quantizing neural networks with the integration of statistical methods in order to track the change in the distribution of weights in the model.

The first phase involves the analysis and testing of appropriate statistical methods designed to compare the distribution of neural network weights. This stage identifies key statistical metrics suitable for monitoring changes in the weight distributions, and performs preliminary tests to detect changes before and after the quantization process is completed. This stage includes the formalisation of a mathematical framework for quantifying changes in weight distributions and the implementation of algorithms for calculating each metric.

The second phase focuses on the implementation analysis and subsequent integration of the selected statistical methods. In particular, the project will focus on three statistical methods, namely: the Kullback-Leibler divergence for asymptotically measuring the distance between the original and the quantized distribution, the Jensen-Shannon divergence as a symmetric metric providing stable results, and the Cosine similarity for measuring changes in the orientation of weight vectors independent of their magnitude.

The third phase implements the design, development and validation of a novel training technique for neural networks that uses statistical comparison of weight distributions. Emphasis is also placed on integrating these methods directly into the learning process. A modular monitoring system for tracking the distributions of weights during training, a logging mechanism and visualization tools are implemented. The proposed solution will be complemented with early termination criteria. Experimental validation is performed on standard datasets (CIFAR-10, CIFAR-100 and others) using different neural network architectures .

The fourth phase implements the testing and evaluation of the design of statistical methods for the neural network training process through experimental validation. The evaluation involves analyzing the performance characteristics of the proposed methods using metrics. The validation process is the testing of the design on different model architectures and application domains to explore the applicability of the developed concept.

DESCRIPTION OF THE ACTIVITIES PERFORMED

The Statistical Approach to Tracking Quantization for Neural Network Training project is on schedule and all set milestones have been met so far. The project started in June 2024 and has successfully implemented the initial activities to select and validate statistical methods suitable for comparing weight distributions in neural networks. The initial phases focused on identifying relevant approaches and pilot testing them on benchmark datasets, allowing to define appropriate procedures for further research. Project activities continued in 2025 and the project moved to the stage of integrating the selected methods into the quantization process in an experimental setting. As part of this integration of the selected statistical approaches into the quantization process, a systematic analysis of their impact on the model metrics during the learning process was also underway.

No significant complications have been observed during the solution so far and the different activities, as part of the work packages, are following each other according to the schedule. Lessons learned have been processed in the form of reports, workshops and publications. At the same time, the outputs are published on the GitHub repository: https://romanbudjac.github.io/samqnn/.

A) WP1 - Analysis and testing of statistical methods

The aim of the first work package was to perform a review analysis and test the statistical methods. An analysis of the properties of statistical methods and their use in a neural network modelling and design environment was performed, aiming to identify existing statistical methods suitable for comparing distributions. Explored existing uses of statistical methods in the field of statistics and their application in neural networks. Three key statistical methods were identified and will be further investigated: the Kullback-Leibler divergence, the Jensen-Shannon divergence, and Cosine similarity.

For each method, a theoretical analysis covering the mathematical foundations, computational complexity and suitability for application in the context of neural networks and their implementation in existing processes is presented.

We created a test environment in the Python programming language using the Keras or Pytorch framework. Three identified statistical methods (Kullback Leibner divergence, Jensen Shanon divergence and Cosine similarity) were implemented with a uniform interface and initial conditions. The underlying datasets were existing publicly available datasets that are standard in the community of researchers and developers. We used the datasets prepared in this way to compute the distributions of weights of neural network distribution models with different characteristics. By using both a real dataset and the distributions obtained by experiments, we matched the real conditions. This avoided the use of synthetic generated data, which would provide the desired outputs but would not account for the realistic conditions of the dataset generation principles. Testing in this way ensures that statistical methods and comparisons of distributions before and after quantization are tested in a uniform manner and guarantees coverage of data points from different types of datasets that correspond to real-world objects.

Testing was performed on the CIFAR10, CIFAR10, MNIST and FMNIST datasets using a simple convolutional network. In order to understand the working of comparing the weight distributions of neural networks on real datasets. For each combination of distributions, the metric values, computation time and stability of the results were recorded. The results obtained were analyzed and the strengths and weaknesses of each method were identified in the context of their use for monitoring quantization. In particular, it was about the impact and load of the added statistical methods in the process of training the neural networks. In Figures 1 and 2 it is possible to vidieť typické rozloženie váh v jednej z vrstiev CNN modelu na datasete MNIST.

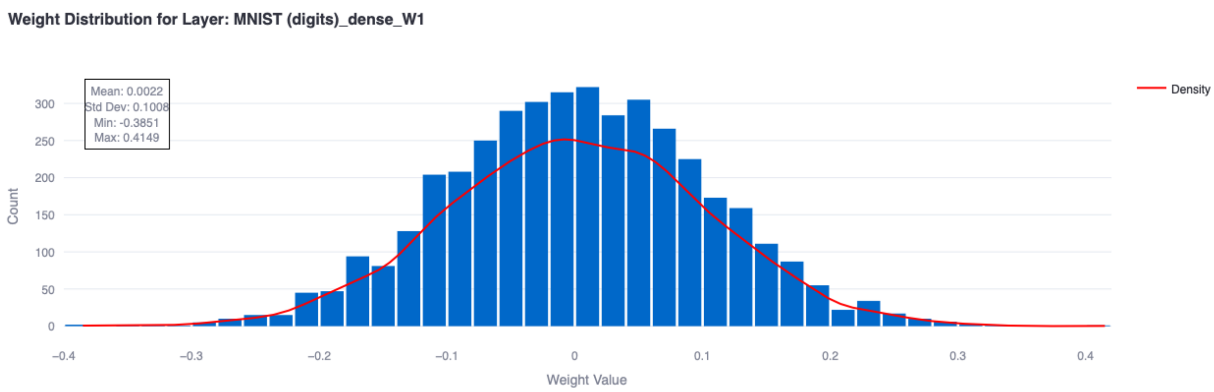

FIGURE 1 Distribution of weights in the dense_W1 layer on the MNIST dataset

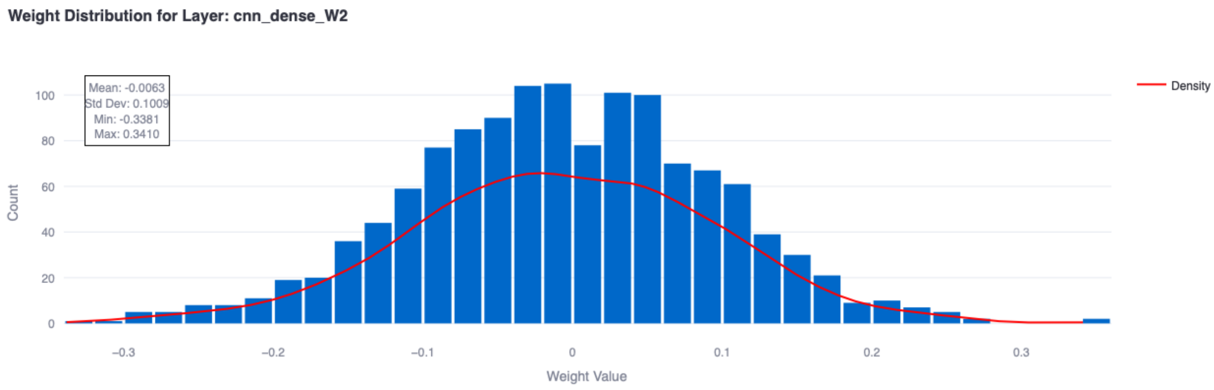

FIGURE 2 Distribution of weights in the dense_W2 layer on the MNIST dataset

Next, in Figure 3 we see the different changes in the distribution of weights before and after quantization. These figures show the distribution of the weights after using 8-bit quantization. In Fig. 3, it is easy to see which parts have been modified by quantization, where blue indicates the original weights and red indicates the frequency of the weights in the original distribution. The area that is overlaid represents the data affected by the quantization process and is no longer found in the data space of the weights matrix of this model. Thus, quantization has reduced the complexity of the distribution of weights in such a way that it approaches the ideal "bell" shape, which is represented in the theoretical analysis by a Gaussian curve.

This effect can be explained by the fact that quantization removes outliers and clusters the weights around the dominant values, thereby narrowing the variance and increasing the spikiness (kurtosis) of the resulting distribution. This transformation is theoretically significant because the Gaussian distribution represents the optimal distribution for a given mean and variance, suggesting that quantization may lead to a more efficient representation of information in the parameter space of the neural network.

In terms of statistical analysis, it can be observed that quantization not only caused a discretization of the values of the weights, but also a change in the fundamental moments of the distribution - the mean, variance, skewness, and skewness. This has important implications for preserving the information capacity of the model and its generalization capability, as reflected in the change in the Kullback-Leibler divergence between the original and quantized weight distributions.

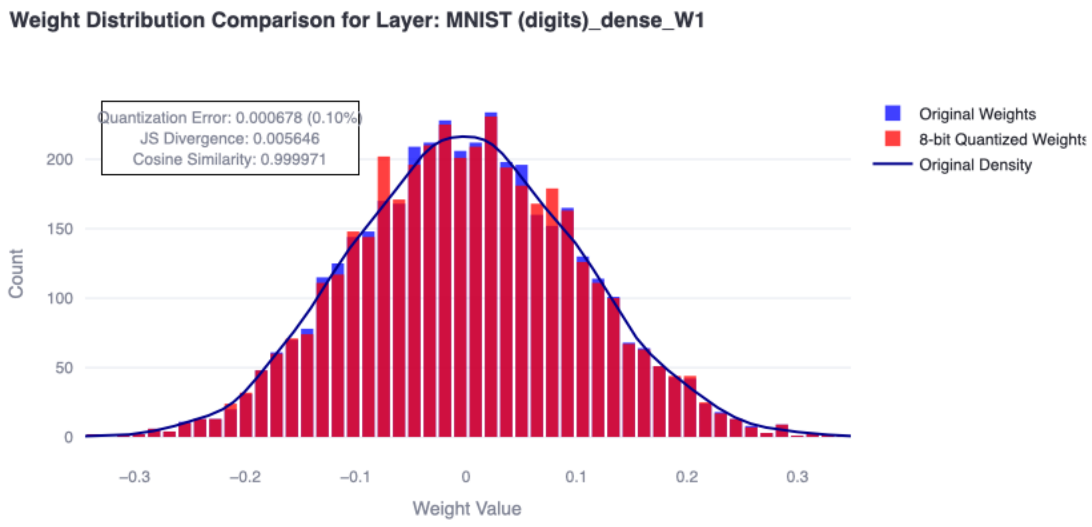

Figure 3 Distribution of weights before and after 8-bit quantization on the MNIST dataset.

In the first work package, an introductory workshop was also organized to present the research problem and the initial state of the project issues.

B) WP2 - Implementation of statistical methods in the quantization process (month 7-12)

As part of the project, a software framework architecture was designed to enable the functional integration of different statistical methods. The algorithm of the system was designed in the form of a functional interface diagram for the individual components. This diagram explains the basic logic of the proposed framework using statistical methods. The architecture has been designed for easy integration into existing training processes.

The implementation was done in Python language using, deep learning framework. Software components were created to extract weights from different types of layers, efficiently calculate distributions and manage reference data. We implemented a custom call mechanism that allowed automatic capture of model state in the form of Checkpoints, which were then used to rebuild the model. In order to reduce the experimental burden of implementing statistical methods in the training process. In this context, vectorized operations with NumPy were used, and the storage of intermediate results was also implemented.

Testing was carried out on different architectures, in particular it was CNN on the MNIST dataset and CNN architectures on the CIFAR-10 dataset. Not only the functionality of calculating changes in the distribution and implementing them in the form of statistical methods was verified, but also the practical impact on the training process. Also, the system of storing checkpoints at different stages of neural network training and their impact was verified The average slowdown of the training process was only 1.8% (in absolute number 0.014s), which confirmed the practical applicability of the solution on the CIFAR10 dataset.

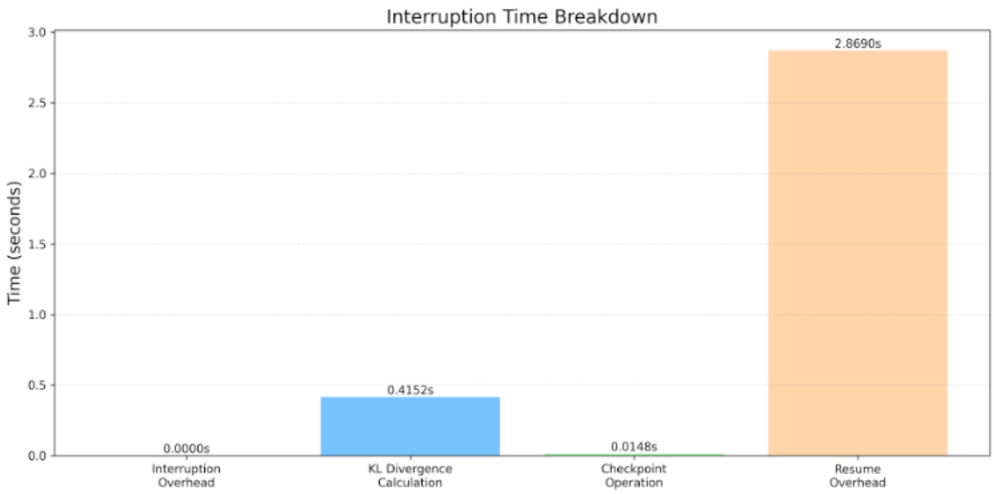

FIGURE 4 Computational complexity of the implementation of statistical methods in the process of quantization and training

As part of the work package, we organised two workshops within the RC Uniza working group, from which we have produced a report. The main benefit of both seminars was to discuss the current status of the project and to hear the expert opinion of the participants.

Dissemination activities in WP1 and WP2, consisted of organising an introductory workshop to present the research problem and the initial state of the project issues. A conference paper was developed from the knowledge gained by implementing statistical methods in the neural network training process at the international conference CSOC 2025 indexed in the SCOPUS database. This was the paper "Layer-wise Statistical Analysis: A Time-Sensitive". Another publication "Proposal for a weather monitoring system utilizing IoT technologies" was presented at the CECIIS 2024 conference in Varaždin was presented at the scientific conference and will be indexed in the SCOPUS database

PROBLEMS, THE RISKS AND THE ACTION TAKEN

During the course of the project, some challenges emerged, but these were identified from the outset within the risk management framework and were therefore addressed through a proactive and predictive approach before they negatively impacted the project. These were mainly :

Risk 1: Risk of inadequate hardware. Probability: medium, Severity: medium.

Measure 1: The host organisation has ensured access to high-performance graphics hardware (GPU) that provides sufficient computing capacity for the experiments carried out as part of the project.

Risk 2: Statistical methods will not be suitable for comparing weights in neural networks. Severity medium, probability low.

Measure 2: We avoided the negative impact of this risk by using several alternative methods for comparing distributions. In case of unsatisfactory results, we continuously modified the methodology based on the data.

Risk 3: Statistical methods for comparing weight distributions do not achieve the expected results in the case of a neural network. Severity: medium Probability: low to medium

Measure 3: We eliminated the negative impact of this risk through an iterative approach in which we continuously evaluated the effectiveness of the methods used and their impact on the metrics of the neural network models.

PROJECT PROGRESS AND IMPLEMENTATION

The research project includes 4 work packages (WP1, WP2, WP3, WP4), these are defined by the four milestones MS1 - MS4.

The first milestone of the project (MS1) involves the analysis of current statistical methods suitable for comparing weight distributions in neural networks and their models. The first project milestone (MS1) involves the analysis and testing of existing statistical methods suitable for comparing weight distributions in neural networks and their models.

The second milestone (MS2) focuses on the analysis of the integration of selected statistical methods directly into the quantization process of neural networks. Within this stage, the possibilities of applying these methods to assess the impact of quantization on the distribution of model weights have been analyzed.

The third milestone (MS3) is aimed at verifying the implementation of the selected statistical methods directly in the neural network training process is embedded statistical approach. This includes the integration of statistical methods into the training process, in order to track changes in the distribution of weights. The result is an implementation that allows evaluating the impact of quantization on individual models.

The fourth milestone (MS4) involves experimental testing of the proposed approach on real data, and final validation of the implemented methods. This includes publication in a scientific journal. The goal of the milestone is to evaluate the concept of implementing statistical methods for the neural network training process.

Evaluation of the achievement of the project objectives:

The first two milestones (MS1-MS2) were successfully met during the project implementation.

At the same time, the project is divided into four work packages (WP1-WP4), each covering key stages of the solution. The first work package (WP1) focuses on the analysis and testing of selected statistical methods for comparing the weight distributions of neural networks. The second work package (WP2) analyses and implements these methods in the context of the quantization process. The implementation of the whole project is defined by four milestones, which reflect the sequence of the different activities of the project.

The objective of the project with regard to the individual work packages in the monitoring period since the start of the project can be summarised as follows.

- WP1: the objective of WP1 was to perform analysis and testing of appropriate statistical methods designed to compare the distribution of neural network weights at different stages of quantization.

- WP2: the aim of WP2 is to analyse and then implement selected statistical methods in the quantization process.

- WP3: the aim of WP3 is to design, develop and validate a new training technique for neural networks that uses statistical comparison of weight distributions, with an emphasis on integrating these methods directly into the learning process and validating them experimentally on standard datasets.

- WP4: The main objective of WP4 is to comprehensively validate the proposed concept

Evaluation of the achievement of the project objectives:

During the implementation of the project, the objectives of the first two work packages (WP1-WP2) were successfully achieved. Analysis and testing of appropriate statistical methods for comparing the distribution of neural network weights was developed. Subsequently, the most appropriate methods were selected and implemented in the software environment and experiments were performed on the datasets in order to compare the impact of quantization on the distribution of weights and the performance of the models. As a next step, the selected statistical methods were successfully integrated directly into the quantization process of the neural networks, thus allowing a detailed analysis of the changes in the distribution of the weights during training and after quantization. All activities were completed in accordance with the project timeline.

Project outputs in the monitoring period:

The following list shows the deliverables (D1-D9) along with the month of delivery of the deliverable according to the project schedule.

1. D1 - Seminar (report) - Month 1
2. D2 - Report Month 3
3. D3 - Source code Month 6
4. D4 - Publication at a scientific conference Month 7
5. D5 - Seminar (Report) Month 8
6. D6 - Report Month 10
7. D7 - Source Code Month 12
8. D8 - Publication at a scientific conference Month 12
9. D9 - Interim Report Month 12

All project deliverables are also delivered on the GitHub repository, in accordance with the project deliverables verification method: https://romanbudjac.github.io/samqnn/

Evaluation of the implementation of the project outputs:

The project outputs were defined in line with the objectives of the individual work packages. During the implementation of the project, outputs D1 to D9 were successfully delivered in accordance with the project schedule. A series of technical workshops were conducted, theoretical analysis and testing of appropriate statistical methods for comparing the distribution of neural network weights were developed and implemented in a modeling environment. The results were presented at the scientific conferences Conference on Information and Intelligent Systems 2024 (CECIIS) and Computer Science On-line Conference 2025, published in an interim report, and all outputs were made available in the project repository. All deliverables were completed in accordance with the project schedule.

PUBLICITY AND COMMUNICATION ACTIVITIES

Dissemination of the project results to the scientific community as well as other relevant target groups takes place through several channels. The results of the project have been presented at international scientific conferences, namely participation in the Conference on Information and Intelligent Systems 2024 (CECIIS) and the Computer Science On-line Conference 2025. Papers from these conferences are (will be) published in proceedings indexed in respected scientific databases (Scopus), thus ensuring a wide reach of the project outputs. Also, double-accented articles have been published in the impacted journals IEEE Access and Applications in Engineering Science published by Elsevier and IEEE. Both in the WOS quartile Q2. The project emphasizes open access principles, so all relevant scientific outputs are made available in a publicly accessible repository, on the GitHub platform.

Currently published publications relevant to the project:

[1] G. Gašpar, R. Budjač, M. Valášek, M. Bartoň, T. Meravý, and M. Strémy, “Digitizing SMEs in the EU: A scalable model for retrofitting machinery to Industry 4.0,” Applications in Engineering Science, vol. 23, p. 100230, Sep. 2025, doi: 10.1016/j.apples.2025.100230.

[2] G. Gaspar et al., “Innovative Grouting Process Monitoring: Design and Implementation of a Measurement Device With Datalogger Function,” IEEE Access, vol. 13, pp. 19340–19352, 2025, doi: 10.1109/access.2025.3534816.

[3] R. Budjac, G. Gašpar, and M. Valášek, "Proposal for a weather monitoring system utilizing IoT technologies," in Proeedings Central European Conf. Information and Intelligent Systems (CECIIS), Varaždin, Croatia, Sep. 2024, pp. 67-74.

V súčasnosti publikácie v recenzii relevantné k projektu:

[4] Budjac R, Valasek M., T. Meravy, and G. Gaspar, "A Multiservice-Oriented Framework for Enhanced Data Collection and Visualization in Smart City Applications," IEEE Access

Currently publications accepted pending release for the project:

[5] R. Budjac and M. Valasek, "Layer-wise Statistical Analysis: A Time-Sensitive Framework for Implementing Statistical Methods in Neural Network Training Sessions," in Lecture Notes in Networks and Systems, Proceedings: *Artificial Intelligence Algorithm Design for Systems*, Springer, [2025], 18p.

DESCRIPTION OF PLANNED ACTIVITIES

In the next period, experiments will continue to develop and implement a training strategy in order to compare the distribution of weights in neural networks before and after the quantization process. Extension of testing to other types of deep neural network architectures using different types of standardized datasets is planned, in order to validate the robustness of the proposed statistical methods in the more general context of AI applications.

The results obtained in the project will be published in a Web of Science indexed scientific journal (Q2 or higher category) and presented at international conferences.